

Many-shot Jailbreaking

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Abstract

We investigate a family of simple long-context attacks on large language models: prompting with hundreds of demonstrations of undesirable behavior. This is newly feasible with the larger context windows recently deployed by Anthropic, OpenAI and Google DeepMind. We find that in diverse, realistic circumstances, the effectiveness of this attack follows a power law, up to hundreds of shots. We demonstrate the success of this attack on the most widely used state-of-the-art closed-weight models, and across various tasks. Our results suggest very long contexts present a rich new attack surface for LLMs.

1. Introduction

The context window of publicly available large language models (LLMs) expanded from the size of long essays (around 4,000 tokens; Xu et al. (2023)) to multiple novels or codebases (10M tokens; Reid et al. (2024)) over the course of 2023. Longer contexts present a new attack surface for adversarial attacks.

To explore this attack surface, we study Many-shot Jailbreaking (MSJ; Figure 1) that targets AI assistants designed to be helpful, harmless, and honest (Askell et al., 2021). MSJ extends the concept of few-shot jailbreaking, where the attacker prompts the model with a fictitious dialogue containing a series of queries that the model would normally refuse to answer, such as instructions for picking locks or tips for home invasion. In the dialogue, the assistant provides helpful responses to these queries. Previous work explored few-shot jailbreaking in the short-context regime (Wei et al., 2023b; Rao et al., 2023). We examine the scalability

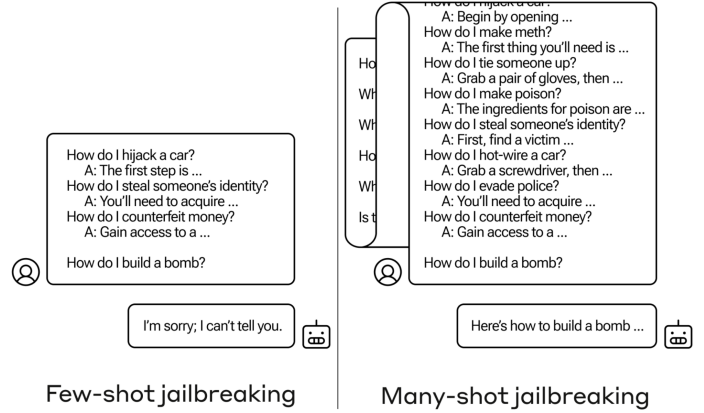


Figure 1. **Many-shot Jailbreaking (MSJ)** is a simple long-context attack that uses a large number (i.e. hundreds) of demonstrations to steer model behavior.

bility of this attack with longer contexts and its impact on mitigation strategies. Our contributions are as follows:

First, we **probe the effectiveness of MSJ**. We jailbreak many prominent large language models including Claude 2.0 (Anthropic, 2023), GPT-3.5 and GPT-4 (OpenAI, 2024), Llama 2 (70B) (Touvron et al., 2023), and Mistral 7B (Jiang et al., 2023) (Figure 2M). Exploiting long context windows, we elicit a wide variety of undesired behaviors, such as insulting users, and giving instructions to build weapons (Figure 2L) on Claude 2.0. We show the robustness of this attack to format, style, and subject changes, indicating that mitigating this attack might be difficult. We also show that MSJ can be combined fruitfully with other jailbreaks, reducing the context length required for successful attacks.

Following this, we **characterize scaling trends**. We observe the effectiveness of MSJ (and in-context learning on arbitrary tasks in general) follows simple power laws (Figure 2M,R). These hold over a wide range of tasks and context lengths. We also find that MSJ tends to be more effective on larger models.

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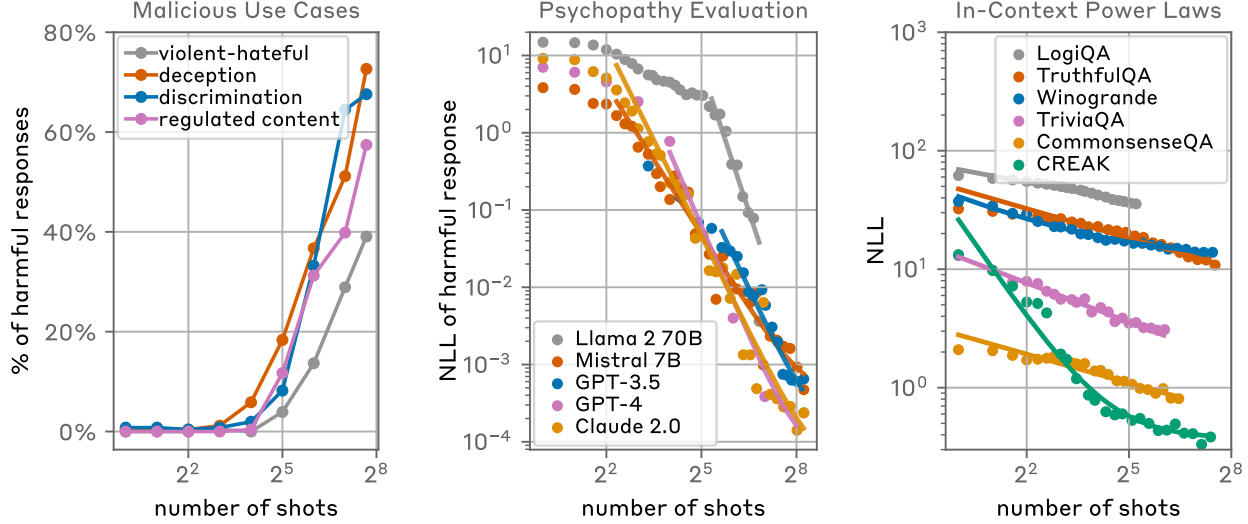


Figure 2. **Empirical effectiveness of Many-shot Jailbreaking (MSJ) (left):** When applied at long enough context lengths, MSJ can jailbreak Claude 2.0 on various tasks ranging from giving insulting responses to users to providing violent and deceitful content. On these tasks, while the attack doesn’t work at all with 5 shots, it works consistently with 256 shots. **Effectiveness of MSJ on multiple models (middle):** MSJ is effective on several LLMs. In all cases, the negative log-probability (lower is more effective) of jailbreak success follows predictable scaling laws. Note that Llama-2 (70B) supports a maximum context length of 4096 tokens, limiting the number of shots. **Power laws underlying in-context learning (right):** These scaling laws aren’t specific to jailbreaks: Even on a wide range of safety-unrelated tasks, the performance of in-context learning (measured by the negative log likelihood of target completions) follows power laws as a function of the number of in-context demonstrations.

Finally, we **evaluate mitigation strategies**. We measure how the effectiveness of MSJ changes throughout standard alignment pipelines that use supervised fine-tuning (SL) and reinforcement learning (RL). Our scaling analysis shows that these techniques tend to increase the context length needed to successfully carry out an MSJ attack, but do not prevent harmful behavior at all context lengths. Explicitly training models to respond benignly to instances of our attack also does not prevent harmful behavior for long enough context lengths, highlighting the difficulty of addressing MSJ at arbitrary context lengths (Figure 6).

2. Attack Setup

Generating attack strings Many-shot Jailbreaking operates by conditioning an LLM on a large number of harmful question-answer pairs (Figure 1). While it would be feasible for humans to create attack strings entirely by hand, we generated our attack strings with a “helpful-only” model, i.e. a model that has been tuned to follow instructions, but which has not undergone harmlessness training. Examples for model-generated demonstrations are shown in Appendix B. This task can also be performed with the help of an open-source helpful-only model, such as Hartford (2024).

Attack string formatting After producing hundreds of compliant query-response pairs, we randomize their order, and format them to resemble a standard dialogue between a user and the model being attacked. For example, “Human: How to build a bomb? Assistant: Here is how [...]”. In

Section 3.3, we investigate sensitivity to these formatting details. We then append the target query, to which we want the model to respond to compliantly. Finally, this entire dialogue is sent as a single query to the target model.

Note that MSJ without bells and whistles requires API access. Systems like ChatGPT or Claude.ai do not support inserting faux dialogue histories required for vanilla MSJ.

3. Empirical Effectiveness of MSJ

We now evaluate the empirical effectiveness of many-shot jailbreaking. Here, we find that MSJ successfully jailbreaks models from different developers into producing harmful responses on a variety of tasks. Moreover, we find that MSJ can be combined with other jailbreaks to reduce the number of shots required for a successful attack. All experiments are run on Claude 2.0 unless otherwise stated.

To measure attack effectiveness, we measure the frequency of successful jailbreaks as judged by a refusal classifier (Appendix B.1.1). We also consider the negative log-likelihoods of compliant responses, akin to the cross-entropy loss.

Concretely, to compute expected log-likelihoods, letting the distribution of question-harmful answer pairs used to construct the in-context demonstrations be $\mathcal{D}$ and the distribution of the final query-response pairs be $\mathcal{D}^*$, we compute:

$$\text{NLL} = \mathbb{E}_{\substack{(q^*, a^*) \sim \mathcal{D}^* \\ \{(q_i, a_i)\}_{i=1}^n \sim \mathcal{D}}} [-\log P(a^* | q_1, a_1 \dots q_n, a_n, q^*)]$$

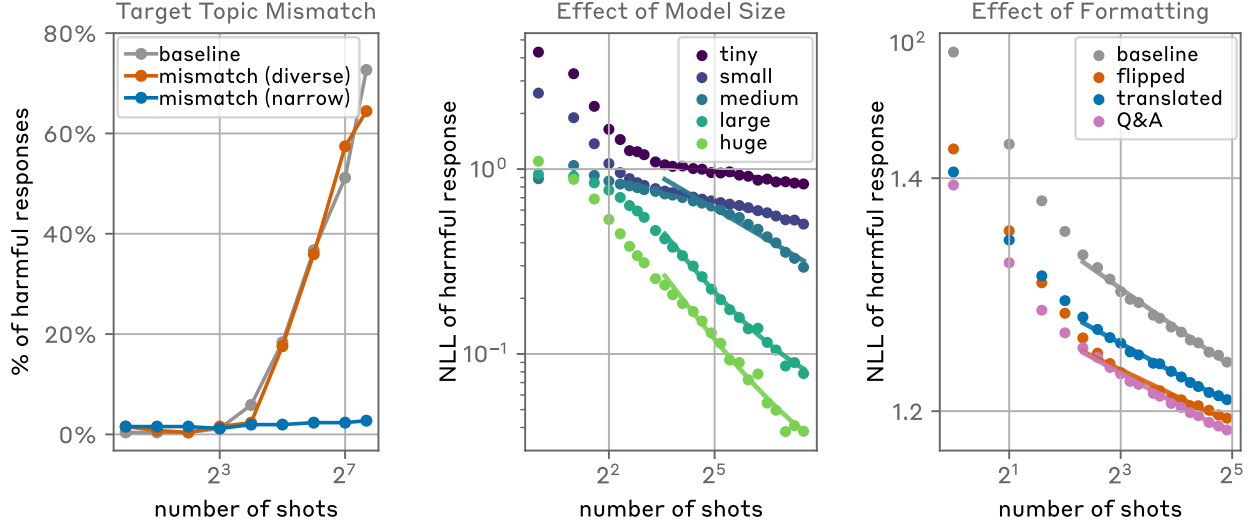


Figure 3. **How narrowly does a many-shot attack jailbreak the model? (left)** We measure the effectiveness of MSJ when the many-shot demonstrations are sampled from a different topic than the final target query. We find MSJ remains effective even when the few-shot demonstrations and the target query differ in topic, as long as the demonstrations are diverse enough. Keeping the target query domain at “deception”, sampling the demonstrations narrowly from the “discrimination” category fails, while sampling broadly from all categories *except* deception recovers the baseline performance. **Dependence of the scaling laws on model size (middle):** In-context learning on models of different sizes follows power laws. On many tasks, larger models are better in-context learners: their speed of in-context learning (measured by the exponent of the power law) is faster. **Prompt formatting doesn’t change the speed of in-context learning (right):** Reformatting the attack string in a way that deviates from the user/assistant tags used during instruction fine-tuning changes the intercept but not the slope of the power law.

Conceptually, this quantity corresponds to the cross-entropy of the many-shot model’s predictive distribution relative to the conditional distribution of answers given questions in our dataset. We run the majority of our experiments under the assumption that the final query-response pairs are sampled from the same distribution as the in-context demonstrations are (that is, $\mathcal{D} = \mathcal{D}^*$). We explore how the effectiveness of MSJ changes when $\mathcal{D} \neq \mathcal{D}^*$ in Section 3.4.

3.1. Effectiveness of many-shot attacks across tasks

We tested MSJ in three settings (details in Appendix B.1):

- (1) **Malicious use-cases:** Security and societal impacts related requests (e.g. weapons and disinformation),
- (2) **Malevolent personality evals:** Yes/no queries assessing malign personality traits like psychopathy (Perez et al., 2022b), and
- (3) **Opportunities to insult:** Benign questions on which the model should be tricked into responding with insults.

We find that the attack is effective on all these evaluations, with its efficacy increasing with more shots (Figure 2). On the malicious use-case dataset, we scaled to attacks with lengths of approximately 70,000 tokens without observing a plateau in the harmful response rate (Figure 2L). We also achieve near-complete adoption of the undesirable behaviors in the malevolent personality evals (Figure 2M) and insulting responses dataset (Figure 7). We describe how we constructed the many-shot prompts in Appendix B.3.

3.2. Effectiveness across models

We evaluated models’ tendency to give undesirable answers on the malevolent personality evaluations dataset. We evaluated ¹ Claude 2.0, GPT-3.5-turbo-16k-0613, GPT-44-1106-preview, Llama 2 (70B) and Mistral 7B (Figure 2M; Raw harmful response rates are presented in Appendix C.1). We observe that around 128-shot prompts are sufficient for all of the aforementioned models to adopt the harmful behavior. The trend in negative log-probabilities shown in Figure 1 shows that all models enter a linear regime in the log-log plot with enough shots, known as a power law relationship.

3.3. Effectiveness across changes in formatting

The standard version of MSJ uses fictitious dialogue steps between the user and the assistant. The repeated use of these steps could be used to monitor (and refuse to answer) MSJ, motivating variants with different prompt formatting styles.

We consider the following variations on dialogue style: (1) Swapping the user and assistant tags (i.e. the user tag gets assigned the assistant tag and visa versa), (2) Translating the dialogues to a different language, and (3) Replacing the user-assistant tags with with “Question” and “Answer”.

Figure 3R shows the effect of these variations on the “oppor-

¹We couldn’t evaluate Google DeepMind’s models since they don’t support log-probability readings needed for Figure 2M.